Calcitriol in the Treatment of Postmenopausal Osteoporosis

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Calcitriol was compared with placebo in the treatment of postmenopausal osteoporosis in a double-blind, randomized, parallel clinical trial of 24 months' duration. Adjustment was made in dietary calcium to maximize the dose of calcitriol. The study was completed by 15 patients who received placebo and 12 patients who received calcitriol. The calcitriol group had positive slopes (compared with negative slopes for the placebo group) for total body calcium, bone mineral content of the radius, bone mineral density of the lumbar spine, and radiographic absorptiometry of the middle phalanges. The difference between the two groups was statistically significant for each of these measurements. The fracture rate in the treatment group was 250 per 1,000 patient-years as compared with 333 for the placebo group. The mean dose of calcitriol was 0.8 μg per day. Hypercalcemia, hypercalciuria, and perhaps nephrolithiasis were observed as complications of treatment. Calcitriol increased bone mineral density by decreasing bone resorption, but not by increasing bone formation. Future studies should concentrate on treatment with oral calcitriol in lower doses. It would also be of interest to examine parenteral administration of calcitriol. It is possible that bone formation can be increased by achieving higher serum levels of the drug, whereas complications may be avoided by using a non-oral route of administration.

Several lines of evidence have implicated decreased circulating levels of calcitriol as a factor in the pathogenesis of postmenopausal osteoporosis. Some investigators have found lower circulating levels of calcitriol in women with the vertebral crush fracture syndrome when compared with age-matched controls [1–3]. There is a decreased synthesis of calcitriol in response to parathyroid hormone infusion in women with postmenopausal osteoporosis [4]. Lower circulating levels of calcitriol are found after the menopause associated with a decrease in calcium absorption. Administration of estrogen or calcitriol to postmenopausal women (with or without the crush fracture syndrome) increases circulating levels of calcitriol to the normal range and restores calcium absorption to premenopausal levels [5,6].

The possibility that calcitriol may stimulate osteoblastic activity as well as increase intestinal calcium absorption suggested that it may be useful in the treatment of postmenopausal osteoporosis [7,8]. Several reports concerning treatment of osteoporosis with calcitriol have appeared in the literature [5,9–11]. The findings have been conflicting, with some reports suggesting that calcitriol is useful whereas others found no benefit from calcitriol treatment. The current study is a 24-month double-blind, randomized, parallel trial designed to examine the safety of calcitriol and to determine if it is more effective than placebo in the treatment of postmenopausal osteoporosis.

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SUBJECTS AND METHODS

Protocol. Recruitment: Women with postmenopausal osteoporosis between the ages of 50 and 80 years were recruited to participate in the study by media releases and letters to physicians. Osteoporosis was diagnosed by the presence of at least one non-traumatic vertebral compression fracture. The women were otherwise healthy; specific exclusion factors included hepatic or renal disease, malignancy, malabsorption syndrome, parathyroid or thyroid disorders, inflammatory arthritis, alcoholism, overt vitamin D deficiency, a history of renal lithiasis, allergy to tetracycline, insulin-dependent diabetes, previous long-term hospitalization, and any other disorder known to affect bone metabolism. Patients were also excluded on the basis of intake of certain drugs, including glucocorticoids, anticonvulsants, estrogens, sodium fluoride, calcium supplements, and pharmacologic doses of vitamin D. Screening consisted of baseline history, physical examination, and routine laboratory chemistry evaluation. Additional studies included a serum thyroxine and serum protein electrophoresis. Prior to initiation of therapy, each patient also had the following performed: assessment of calcium and hydroxyproline in a 24-hour urine collection obtained after three days of ingesting a low hydroxyproline diet, determination of creatinine clearance, measurement of serum levels of parathyroid hormone (PTH) and vitamin D metabolites, bone mineral measurements, eye examination by an ophthalmologist, and anterior iliac crest bone biopsy.

Telephone contact was made with 883 women who responded to the recruitment effort; of those interested in participating in the study, 91 were not excluded by telephone interview. Of these 91 women who were screened, 34 were included in the study. The most common reason for exclusion of the patients who were examined was the absence of a vertebral crush fracture on radiography. Sample size was determined by knowledge of average rates of bone loss, the variation in the bone mineral measurements, the number of bone mineral measurements and the interval between them, and the assumption of a dropout rate of 15 percent.

Dose titration: On entry into the study, each patient received 400 IU of vitamin D daily. Dietary calcium intake was assessed by a three-day diet history and patients were instructed in a 1,000-mg calcium intake. All patients received two capsules per day, one before breakfast and the second before dinner. The capsules contained either 0.25 μg of calcitriol or a matching placebo. The following stepwise escalation in dose was permitted: after two weeks to two capsules twice daily, at one month to three capsules twice daily, and after six weeks to the maximum of four capsules twice daily. The investigator also had the option of stepwise escalation of one capsule instead of two every two weeks. The dosage schedule was to be modified to control hypercalciuria (greater than 400 mg of calcium per 24-hour urine sample), if lowering dietary calcium intake was ineffective. If significant hypercalcemia developed, greater than 11.6 mg/dl), the drug was to be temporarily discontinued and the dose was initiated later at one capsule daily. Informed consent was obtained from each patient.

Continuation phase: Following completion of 24 months of study, the code was broken. All participants were invited to continue in an open study to assess long-term safety. The calcitriol group continued to receive the same dose as previously and the placebo group started to receive calcitriol, using the same titration protocol as during the blinded phase of the study. Eighteen patients agreed to continue in the study, six who previously received calcitriol and 12 who previously received placebo. At the time the study was ended, the mean additional length of time in the continuation phase was 18.4 ± 1.9 (SE) months for those who previously received calcitriol and 16.7 ± 1.5 months for those who previously received placebo. At the completion of this continuation phase of the study, it was decided to perform renal sonograms on each patient; 14 patients agreed to have sonography.

Methods. Bone mineral measurements: Bone mineral measurements were performed at baseline and every six months, using previously described techniques [12–14]. Total body calcium, which essentially provides a measure of calcium balance in a free-living population, was measured in the Brookhaven National Laboratory facility by total body neutron activation analysis and whole body counting. The reproducibility of this technique is ±1 percent in an anthropomorphic phantom and ±2 to 3 percent in postmenopausal women. Bone density of the distal radius (8-cm site) was measured by single photon absorptiometry using a Norland Instruments (Ft. Atkinson, Wisconsin) absorptiometer with an ^{125}I source. Bone density of the lumbar spine (L2 to L4) was measured using dual photon absorptiometry with a ^{153}Gd source (Lunar Instruments, DP_3, Madison, Wisconsin). The reproducibility of these bone density measurements in normal postmenopausal women is approximately ±2 to 3 percent. Radiographic absorptiometry of the second, third, and fourth middle phalanges of the left hand was performed using scanning with an optical densitometer and computer for detection of bone edges. This technique has a reproducibility of ±1.6 percent [15].

Biochemical determinations and radiographs: Urinary hydroxyproline was determined by a modification of the method of Prockop and Udenfriend [16]. Vitamin D metabolites were measured by the technique of Eisman et al [17]. PTH radioimmunoassay employed a Diagnostic System Laboratories radioimmunoassay kit. The antiserum in this assay is directed against the C-terminal PTH molecule. In order to reduce interassay variability, sera for PTH and vitamin D assays were frozen and stored at −30°C so that the assays from all treatment periods could be determined simultaneously. Calcium was measured on an atomic absorption spectrometer and phosphate was measured by the technique of Fiske and Subbarow [18]. Serum samples were collected after an overnight fast. Following completion of the study, it was decided to measure serum levels of osteocalcin [19]. Samples were available on 11 patients in the treatment group for the following approximate time limits: baseline, three, six, nine, and 12 months. All samples were assayed simultaneously.

Radiographs were interpreted by the Brookhaven National Laboratory radiologist to determine eligibility for inclusion in the study. Subsequently, fracture incidence was determined by analysis of pretreatment and annual radiographs of the thoracolumbar spine by one physician who

was unaware as to whether the patients were receiving placebo or calcitriol.

Bone biospy specimens: A transiliac specimen was obtained with a Bordier trephine under local anesthesia for quantitative histomorphometric analysis [20]. Double tetracycline labeling was utilized; the biopsies were performed at baseline and upon completion of the study. All patients had baseline biopsies; repeat biopsies were performed on 12 patients in the placebo group and 10 patients in the calcitriol group. All biopsies were interpreted by one physician who was unaware of the treatment received by the patients.

The parameters reported are defined as follows: trabecular bone volume (mm^3/mm^3): the area or volume of marrow space occupied by bone tissue; fractional osteoid volume (mm^3/mm^3): the portion of the area or volume of trabecular bone tissue that consists of unmineralized osteoid; osteoid volume (mm^3/mm^3): the portion of the area or volume of marrow space occupied by unmineralized osteoid; appositional rate (mm/year): the rate of advancement of the bone-forming front taken from the tetracycline labels and the marker interval; osteoid seam thickness (mm): the average thickness of osteoid seams corrected for obliquity; formation surface (mm^2/mm^2): the fraction of total trabecular surface occupied by unmineralized osteoid; resorption surface (mm^2/mm^2): the fraction of total trabecular surface occupied by Howship's lacunae, with and without osteoclasts; bone formation rate total surface referent (mm^3/mm^2/year): the volume of mineralized new bone made per unit of total trabecular surface per unit time; mineralization lag time (years): the average length of time between the creation of osteoid matrix and its mineralization.

Statistical analysis: Individual slopes of each of the bone mineral measurements were computed and expressed as percent per year; each data point was expressed as the difference from the mean of all of the individual values. The slopes of the calcitriol and control groups were then compared by the student t test to determine if calcitriol was more effective than placebo. Since the hypothesis was unidirectional, a one-tailed t test was used. The data presented are for the subjects that completed the 24 months of the study. In addition, slopes were computed to include the patients who dropped out of the study; for this latter analysis, the slopes were weighted for the number of measurements and duration in the study. A multivariate analysis of variance was carried out using each of the four different bone mineral measurements (weighted appropriately and expressed as an annual rate of change) as criteria. The biochemical data that were obtained each six months were analyzed using one-way analysis of variance. The two groups were compared using an unpaired t test. Analyses were carried out using the SPSS-x21. The fracture rate was tested for significance using the Mann-Whitney test. The histomorphometry data were evaluated by the paired t test.

RESULTS

Subjects. The mean age of the placebo group was 64.9 ± 1.7 (SE) years and that of the calcitriol group was 64.1 ± 1.5 (SE) years (**Table I**). Other clinical characteristics were similar for the two groups. Seven patients dropped out of the study. The reasons for the dropouts were not complications of treatment as far as could be determined. The two dropouts in the placebo group occurred because one woman had a myocardial infarction and another believed that dizziness and nausea were due to the "drug." Dropouts from the calcitriol group occurred for the following reasons: myocardial infarction, psychiatric disturbance, death of a spouse, change in residence, and discouragement when a new fracture occurred.

TABLE I Baseline Characteristics (mean ± SE)

	Calcitriol (n = 12)	Placebo (n = 15)	p Values
Age	64.1	64.9	NS
	1.5	1.7	
Age at menopause	46.8	47.2	NS
	1.9	1.6	
Number of years postmenopausal	17.3	17.6	NS
	2.1	2.7	
Number of children	2.2	2.3	NS
	0.4	0.4	
Smoking history	5.5	12.1	NS
	3.6	4.2	
Fractures	2.50	3.20	NS
	0.39	0.63	

Dosage and Complications. The mean dose of calcitriol was 0.8 μg per day. The mean number of capsules for the placebo group was 7.6, indicating adherence of the investigators to the protocol in dosage titration. Dietary calcium was lowered in each of the calcitriol-treated patients by 250 mg to 500 mg per day because of persistent hypercalciuria. In the titration phase, the dosage ranged from 1 to 2 μg per day, before a reduction in dosage was necessary. The daily dose of calcitriol following titration was 0.5 to 0.75 μg for eight subjects, with four subjects taking 1 to 1.25 μg per day. After dietary modification, the serum calcium levels returned to the normal range, but remained higher than the values in the control subjects. Hypercalciuria occurred in all subjects treated with calcitriol. Hypercalciuria preceded the hypercalcemia by about two weeks. During the titration phase, hypercalcemia (greater than 11 mg/dl) occurred in 11 of the 12 subjects receiving calcitriol. The maximal elevation in serum calcium was seen in one subject who was given a thiazide diuretic for hypertension by her physician. Her serum calcium level increased to 14.0 mg/dl, and she required intravenous saline therapy along with discontinuation of the calcitriol and the thiazide to bring the hypercalcemia under control. No abnormalities were noted on ophthalmologic examination. Renal sonography revealed a stone in one patient. This patient had taken calcitriol for 31 months. No evidence of nephrocalcinosis was found on the sonograms. Although there was no change in mean values for creatinine clearance, a clear decrease in creatinine clearance was observed in two patients taking calcitriol; one of these patients was the person in whom nephrolithiasis was noted on sonography.

TABLE II Initial Values and Rate of Change for Bone Mineral Measurements (mean ± SE)

	Number	Total Body Calcium (g)	Total Body Calcium (percent/year)	Bone Mineral Content Radius (g/cm)	Bone Mineral Content Radius (percent/year)	Bone Density Spine (g/cm)	Bone Density Spine (percent/year)	Radiographic Absorptiometry (μ)	Radiographic Absorptiometry (percent/year)
Calcitriol	12	650 ± 34	0.967* ± 0.706	0.663 ± 0.040	1.26† ± 1.03	0.878 ± 0.055	0.18* ± 1.80	81.70 ± 4.45	0.98* ± 0.56
Placebo	15	618 ± 19	−0.73 ± 0.55	0.665 ± 0.032	−1.63 ± 0.59	0.849 ± 0.044	−4.27 ± 1.00	81.30 ± 3.30	−1.39 ± 1.22

* $p < 0.05$.
† $p < 0.01$.

Bone Mineral Measurements. There were no significant differences between the two groups in the baseline bone mineral measurements (**Table II**). The mean change in the bone mineral content of the radius for the calcitriol group was 1.26 per year as contrasted with a change of −1.63 percent per year for the placebo group ($p < 0.01$) (Table II). There was no loss of bone density of the spine in the calcitriol group (0.18 percent per year) whereas a loss of 4.3 percent per year was observed in the placebo group ($p < 0.05$). The same trend was observed in overall calcium balance as estimated from the total body calcium (+0.96 percent per year for the calcitriol group and −0.73 percent per year for the placebo group; $p < 0.05$). The following slopes were computed for the radiographic absorptiometry measurements: 0.98 percent per year for the calcitriol group and −1.39 percent per year for the placebo group ($p < 0.03$). One of the patients in the placebo group experienced a marked reduction in this radiographic measurement (**Figure 1**). When the data were analyzed with this patient's data omitted, the difference between the two groups remained statistically significant ($p < 0.05$).

No relationship was found between the dose of calcitriol and the individual rates of changes in bone mineral measurements. When the data were analyzed to include the patients who dropped out, statistically significant differences for each of the bone mineral measurements persisted except for the total body calcium measurement (bone mineral content of the radius: $p < 0.01$; bone density of the spine: $p < 0.01$; radiographic absorptiometry: $p < 0.05$; and total body calcium: $p = 0.15$). The multivariate analysis of variance also indicated a significant treatment effect and Hotelling's multivariate $F = 2.82$ ($p < 0.01$).

Fracture Incidence. There were more fractures in the placebo group at baseline (66 [range, one to 10]) than in the treatment group (32 [range, one to seven]). The mean number of compression fractures for the placebo and calcitriol groups were 3.2 ± 0.6 (SE) and 2.5 ± 0.4, respectively. During the 24 months of the study, 10 new fractures occurred in five placebo-treated patients and six new fractures occurred in three calcitriol-treated patients, giving a fracture rate (per 1,000 patient-years) of 333 and 250 for the placebo and calcitriol groups, respectively. The difference in fracture incidence between the two groups was not statistically significant.

Biochemical Measurements. The significant changes in the biochemical indices were confined to the treatment group. There were no significant baseline differences between the two groups in the biochemical variables (**Table III**). One-way analysis of variance revealed significant changes with time in the calcitriol group as follows: urinary calcium increased ($F = 4.18$, $p < 0.01$) and there was a decrease in urinary hydroxyproline ($F = 2.64$, $p < 0.05$) and serum alkaline phosphatase ($F = 2.65$, $p < 0.05$). There was no change in serum osteocalcin levels. Mean (±SE) values for serum osteocalcin (ng/ml) were as follows: baseline: 9.04 ± 1.6; three months: 11.0 ± 0.91; six months: 11.5 ± 2.7; nine months: 8.94 ± 1.3; and 12 months: 8.95 ± 1.3 ($F = 0.38$). There were no changes in mean PTH levels or in creatinine clearance.

Bone Biopsies (Table IV). The trabecular bone volume was below 20 percent of the marrow space as would be expected in osteoporotic patients. The values for the osteoid volume, osteoid seam width, appositional rate, and active formation surface were all consistent with the absence of osteomalacia on the pretreatment biopsy specimens [20]. There were no significant changes between the two groups or within each group following treatment (t test). Comparison of the baseline values of those who did not have a second biopsy with those who had two biopsies revealed no significant difference between the dropouts and those patients that completed the study.

COMMENTS

Calcitriol was observed to be effective in the prevention of bone loss in women with postmenopausal osteoporosis. Calcium balance (as assessed by measurement of total body calcium) was positive in the group treated with calcitriol and was negative in the placebo group. Similar findings in the density of the radius and metacarpals confirms that the calcium is deposited in bone rather than in extraskeletal tissues. Moreover, the density of the spine decreased in the placebo group, whereas it did not change in the treatment group.

Several previous studies concerning treatment of postmenopausal osteoporosis with calcitriol have been reported, but few of these measured bone mass or density. Caniggia et al [21] studied calcitriol treatment in both an

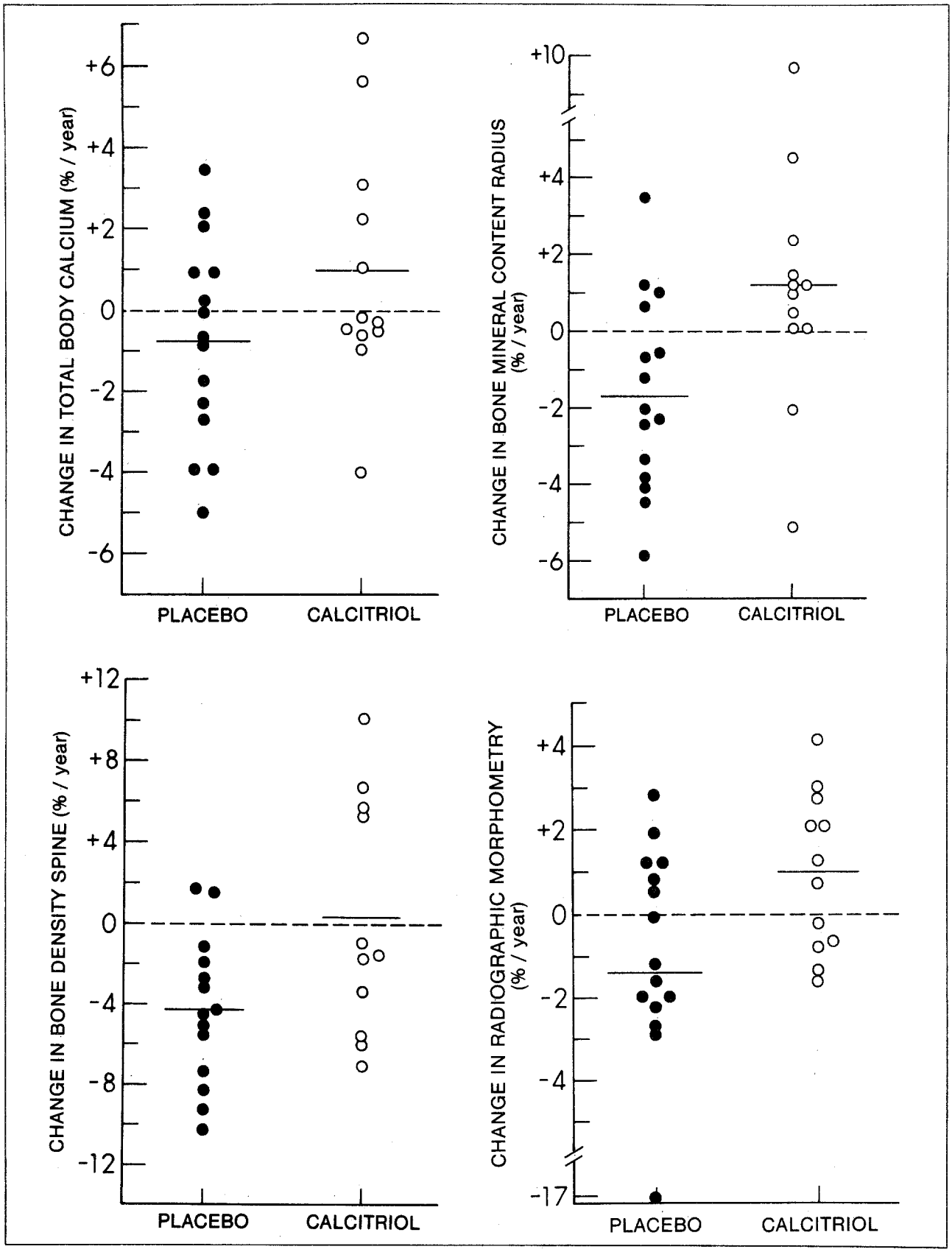


Figure 1. *Bone mineral measurements are expressed as percent per year for each subject who completed the 24 months of study.*

TABLE III Baseline Values and Changes Over Time (in percent from baseline) in Biochemical Studies

		Baseline Value	6 months (percent)	12 months (percent)	18 months (percent)	24 months (percent)
Serum calcium (mg/dl)	Calcitriol	9.43 ± 0.13	103.9 ± 1.80*	102.2 ± 1.86	103.1 ± 1.72	102.8 ± 2.01
	Placebo	9.36 ± 0.13	99.3 ± 1.04	99.3 ± 0.87	99.9 ± 0.83	99.8 ± 1.03
Urinary calcium (mg/g creatinine)	Calcitriol	235 ± 71	181 ± 24.4	209 ± 26.1†	197 ± 32.4*	148.8 ± 23.4
	Placebo	141 ± 13	140 ± 25.9	112 ± 9.35	119 ± 13.9	163 ± 34.2
Creatinine clearance (ml/minute)	Calcitriol	69.4 ± 6.9	96.9 ± 14.1	87.5 ± 9.9	96.4 ± 9.9	101 ± 12.0
	Placebo	58.7 ± 6.1	131 ± 18.8	124 ± 19.2	107 ± 11.1	125 ± 21.4
Urinary hydroxyproline (mg/g creatinine)	Calcitriol	37.8 ± 15.6	89.6 ± 18.8	89.0 ± 13.4	78.7 ± 15.6	76.9 ± 10.3
	Placebo	29.5 ± 2.92	116 ± 15.0	101 ± 13.0	103 ± 16.5	102 ± 15.1
Serum alkaline phosphatase (IU/liter)	Calcitriol	79.3 ± 4.23	84.5 ± 3.58	80.9 ± 3.57	83.3 ± 39.3	82.3 ± 4.54
	Placebo	96.4 ± 9.17	88.4 ± 7.51	75.6 ± 6.02	74.8 ± 6.91	80.3 ± 5.92
Serum PTH (pg/ml)	Calcitriol	591 ± 58	97.5 ± 8.15	94.6 ± 7.34	101 ± 4.33	109 ± 4.86
	Placebo	550 ± 36	102 ± 6.24	102 ± 5.89	101 ± 4.90	103 ± 6.20
Serum 1,25-dihydroxyvitamin D (pg/ml)	Calcitriol	31.5 ± 3.3	154 ± 17.4*	140 ± 19.6	161 ± 17.4†	164 ± 21.8*
	Placebo	32.2 ± 3.1	110 ± 8.3	116 ± 12.3	108 ± 7.8	108 ± 10.8
Serum 24,25-dihydroxyvitamin D (ng/ml)	Calcitriol	1.76 ± 1.03	106 ± 20.2	146 ± 66.8	137 ± 38.7	94.0 ± 18.9
	Placebo	1.45 ± 0.86	114 ± 18.6	69.6 ± 12.2	123 ± 28.7	82.0 ± 15.4
Serum 25-hydroxyvitamin D (ng/ml)	Calcitriol	21.9 ± 7.12	122 ± 18.0	101 ± 12.7	120 ± 24.9	112 ± 18.8
	Placebo	26.6 ± 11.7	114 ± 11.1	87.2 ± 15.1	98.3 ± 11.4	87.9 ± 11.3

* Significantly different from the placebo group at the respective time period by the Student t test ($p < 0.05$).
† Significantly different from the placebo group at the respective time period by the Student t test ($p < 0.01$).

TABLE IV Histomorphometry of Anterior Iliac Crest Biopsies

	Calcitriol			Placebo		
Bone Biopsy Parameters	Initial Biopsy (Mean ± SE)	Final Biopsy (Mean ± SE)	t Value	Initial Biopsy (Mean ± SE)	Final Biopsy (Mean ± SE)	t Value
Trabecular bone volume (mm^3/mm^3)	0.1750 ± 0.0328	0.1131 ± 0.0136	2.03	0.1480 ± 0.0138	0.1210 ± 0.0191	1.19
Fractional osteoid volume (mm^3/mm^3)	0.0113 ± 0.0023	0.0196 ± 0.0065	1.44	0.0121 ± 0.0028	0.0307 ± 0.0111	1.76
Osteoid volume (mm^3/mm^3)	0.0018 ± 0.0004	0.0026 ± 0.0009	0.89	0.0017 ± 0.0005	0.0038 ± 0.0013	1.89
Osteoid seam width (mm)	0.0074 ± 0.0007	0.0075 ± 0.0008	0.07	0.0090 ± 0.0006	0.0087 ± 0.0005	0.91
Fractional resorption surface (mm^2/mm^2)	0.0348 ± 0.0076	0.0466 ± 0.0207	0.06	0.0550 ± 0.0064	0.0553 ± 0.0127	0.17
Appositional rate (mm/year)	0.1749 ± 0.0091	0.1487 ± 0.0180	1.05	0.1803 ± 0.0085	0.1945 ± 0.0106	0.99
Fractional active formation surface (mm^2/mm^2)	0.0353 ± 0.0096	0.0400 ± 0.0137	0.04	0.0549 ± 0.0106	0.0672 ± 0.0162	0.77
Mineralization lag time (years)	0.2507 ± 0.0604	0.1645 ± 0.0370	1.23	0.1936 ± 0.0311	0.1467 ± 0.0266	2.18
Bone formation rate, total surface referent (mm^3/mm^2/year)	0.0338 ± 0.0074	0.0566 ± 0.0148	0.96	0.0558 ± 0.0082	0.0760 ± 0.0120	2.19

open and doubly blinded protocol. They observed a statistically insignificant increase in bone density of the radius in a group treated with 0.5 μg of calcitriol per day. Jensen et al [9] treated 70-year-old women with calcitriol or estrogen/gestagen and noted that the bone mineral content of the radius increased in the groups treated with estrogen/gestagen, but remained unchanged or decreased with calcitriol therapy. The reason for the discordance with the current study is uncertain but may be due to differences between elderly women and women with vertebral crush fractures.

If calcitriol influenced bone remodeling, the changes were too small to be detected in samples of this size with a technique having the measurement reproducibility of histomorphometry. The lack of change in the bone biopsy measurements does indicate, however, that treatment with calcitriol does not produce the dramatic changes in bone remodeling that would be necessary to restore the

skeleton to a non-osteoporotic state. Moreover, since the efficacy phase of this study was conducted for only 24 months, it is not known whether calcitriol treatment will produce a sustained increment in bone mass, or whether there will be a "plateau effect," as has been observed with other drugs that reduce bone remodeling such as calcitonin or estrogens [22,23].

It is theoretically possible to increase bone mass or density without increasing bone strength if the quality of the new bone is deficient. In the current study, there were fewer new fractures in the treatment group compared with the number in control group at the end of the 24 months of treatment. The number of new fractures in the calcitriol groups may be cautiously compared with other reports of fracture rate and suggests a beneficial effect of treatment [24]. We are reluctant to emphasize this finding because of the small number of new fractures in a limited patient sample; however, there certainly was no evidence that calcitriol reduced the quality of bone. It should also be noted that the placebo group in the current study had more fractures at baseline. It is not known whether this difference could have influenced the outcome of our study.

The pretreatment serum calcitriol levels in our patients were similar to those observed in women of the same age without fractures. In a previous study, we observed slightly lower calcitriol levels in osteoporotic women compared with age-matched normal subjects, but these lower levels were proportional to their slightly reduced levels of 25-hydroxyvitamin D [3]. Our current patient population did not have reduced levels of 25-hydroxyvitamin D, which may explain the difference in calcitriol levels. At any rate, it is clear from the serum levels of calcitriol following treatment that the current study achieved pharmacologic rather than replacement therapy drug levels.

Although intestinal calcium absorption was not directly measured in this study, the universal increase in urinary calcium excretion with an associated increase in bone mass indicates that calcium absorption did increase in response to treatment. Calcitriol produced positive calcium balance by a reduction in bone resorption (as evidenced by reduced urinary hydroxyproline) accompanied by an increase in calcium absorption. In addition to the current study, a decrease in hydroxyproline excretion following calcitriol therapy was noted by Riggs and Nelson [10], Need et al [5], and Gallagher et al [11]. It is possible that bone resorption decreased as a result of reduced PTH secretion in response to an increase in serum calcium. We did not detect a change in serum PTH levels. However, the PTH assay we used does not have sufficient sensitivity to detect small changes in PTH secretion. In addition, a radioimmunoassay directed against the biologically active N-terminal of the PTH molecule might have detected changes in PTH secretion.

The protocol in the current study was designed to maximize serum levels of calcitriol in the hope that there would be an effect on osteoblastic proliferation in addition to a reduction in bone resorption. A major effect on bone formation probably did not take place. Urinary hydroxyproline excretion decreased by one third, which reflects a reduction in bone resorption sufficient to account for the modest increments we observed in bone mass. Furthermore, serum alkaline phosphatase levels decreased following treatment, and there was no evidence of increased bone formation on histologic examination of bone. Finally, osteocalcin levels did not increase. The discordance between our findings and previous reports concerning osteocalcin levels could be due to differences between acute and chronic effects of calcitriol administration. However, Caniggia et al [21] recently reported an increase in serum osteocalcin levels following long-term treatment with calcitriol.

The question may also be raised as to whether there is any advantange of calcitriol treatment over calcium supplementation in overcoming the reduced calcium efficiency in women with postmenopausal osteoporosis. Other studies suggest that many women could not achieve zero calcium balance even if given oral calcium supplementation; it is also established that the administration of calcitriol regularly achieves increased absorption efficiency of calcium [11,25,26]. Furthermore, we have previously observed that osteoporotic women who are given 1,000 mg per day of calcium supplementation do not achieve zero calcium balance [27]. In the current study, the control group had a calcium intake of 1,000 mg per day and received a vitamin D supplement of 400 IU daily. This regimen failed to prevent bone loss. Thus, there may be potential benefit in increasing calcium absorption efficiency through administration of calcitriol. Future studies should evaluate the benefit/risk of oral calcitriol in lower doses (0.25 to 0.5 μg).

The average dose of calcitriol used in the current study was 0.8 μg per day. Concomitant prescription (or self-administration) of calcium supplements could result in hypercalcemia at this dosage. Riggs and Nelson [10] treated 20 women with postmenopausal osteoporosis with calcitriol in a dose of 0.25 μg twice daily (n = 12) or three times a day (N = 18). The dosage was reduced to 0.5 μg daily in 13 patients and in seven patients to less than 0.5 μg per day, based on the development of mild hypercalcemia [7], hypercalciuria greater than 350 mg per day [8], and a relative increase in urinary calcium in relationship to glomerular filtration rate [5]. The marked hypercalcemia we observed in one patient in the current study following prescription of a thiazide diuretic by another physician underscores the possible toxic effects of treatment with calcitriol. The possible development of a kidney stone and reduced creatinine clearance in one patient is disconcerting, as was a reduction in creatinine clearance in one other patient.

In conclusion, calcitriol treatment reduces bone loss in women with postmenopausal osteoporosis by increasing intestinal calcium absorption and reducing bone resorp-

tion. We found no evidence for calcitriol having an effect on bone formation. The development of hypercalciuria and hypercalcemia is dose related, whereas an increase in calcium absorption can be attained with lower doses than used in the current study. We do not believe that calcitriol should be used to treat postmenopausal osteoporosis in the higher oral doses employed in this study. Further studies of efficacy and safety with lower doses of calcitriol (up to 0.5 μg per day) would be of interest. Moreover, the side effects we observed were in part due to the oral route of administration, i.e., absorptive hypercalciuria resulted from the intestinal passage of calcitriol. It is likely that higher levels of serum calcitriol could be achieved using a parenteral route of administration without the side effects that we observed. Thus, further studies achieving similar or even higher serum levels of calcitriol by parenteral administration of the drug would also be of interest.

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